

Fundamentals of Information Security

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Transposition Cipher

- A *transposition cipher* rearranges the characters in the plaintext to form the ciphertext. The letters are not changed.

EXAMPLE: The *rail fence* cipher is composed by writing the plaintext in two rows, proceeding down, then across, and reading the ciphertext across, then down. For example, the plaintext “HELLO, WORLD” would be written as:

HLOOL
ELWRD

resulting in the ciphertext “HLOOLELWRD.”

- How will “ELEMENTARY WATSON” be written under transposition cipher ?

Columnar Transposition Cipher

- Columnar Transposition involves writing the plaintext out in rows, and then reading the ciphertext off in columns one by one
- The message is written out in rows of a fixed length, and then read out again column by column, and the columns are chosen in some scrambled order.
- Width of the rows and the permutation of the columns are usually defined by a **keyword**.
- For example, the word HACK is of length 4 (so the rows are of length 4), and the permutation is defined by the alphabetical order of the letters in the keyword. In this case, the order would be “3 1 2 4”.

Columnar Transposition Cipher

- Any spare spaces are filled with nulls or left blank or placed by a character (Example: _).
- Finally, the message is read off in columns, in the order specified by the keyword.

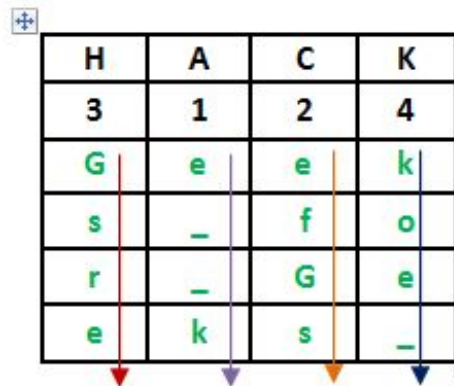
Encryption

Given text = Geeks for Geeks

Keyword = HACK

Length of Keyword = 4 (no of rows)

Order of Alphabets in HACK = 3124



H	A	C	K
3	1	2	4
G	e	e	k
s	_	f	o
r	_	G	e
e	k	s	_

Print Characters of column 1,2,3,4

Encrypted Text = e kefGsGsrekeo_

Columnar Transposition Cipher

- To decipher it, the recipient has to work out the column lengths by dividing the message length by the key length.
- Then, write the message out in columns again, then re-order the columns by reforming the key word.